

MACHINE LEARNING APPLICATIONS IN FINANCIAL LEASING

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Project Proposal Report

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i. List of Abbreviations

Table 1.1 List of Abbreviations

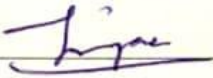
Abbreviation	Full Form
ML	Machine Learning
AI	Artificial Intelligence
IFRS 9	International Financial Reporting Standard 9
ECL	Expected Credit Loss
LGD	Loss Given Default
PD	Probability of Default
EL	Expected Loss
ROC	Receiver Operating Characteristic
AUC	Area Under Curve
RMSE	Root Mean Square Error
MSE	Mean Squared Error
KPI	Key Performance Indicator
CSV	Comma-Separated Values
GPU	Graphics Processing Unit
API	Application Programming Interface
JSON	JavaScript Object Notation

ii. Declaration of The Candidate

I declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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The supervisor/s should certify the proposal report with the following declaration. The above candidates are carrying out research for the undergraduate Dissertation under my supervision.



Signature of the supervisor

27/08/2025

Date



Signature of the Co-supervisor

27/08/2025

Date

iii. Abstract

This research proposal explores the application of machine learning techniques in the financial leasing sector, focusing on impairment analysis. Impairment occurs when the expected recoverable amount of leased financial assets decreases, leading to significant financial risk. Traditional statistical models struggle with the dynamic and non-linear nature of financial data [1], [2]. This study proposes a machine learning-based system for early detection and prediction of impairment events in leasing portfolios [3], [4], [5]. The system aims to improve decision-making, minimize credit risk, and align with IFRS 9 expected credit loss (ECL) standards [6], [7].

The suggested system will examine past lease data, financial indicators, and lessee behavior patterns using a mix of supervised and unsupervised machine learning approaches. The methodology seeks to offer a more precise and prompt evaluation of any impairments by seeing minute patterns and correlations that conventional approaches frequently miss. Leasing firms may minimize unforeseen financial losses, optimize provisioning techniques, and prioritize high-risk assets with this proactive approach.

The study will also concentrate on creating an interpretable and flexible framework. By providing explicit insights into the causes influencing impairment estimates, the system aims to assist decision-makers, improve stakeholder confidence, and facilitate greater regulatory compliance. The method can adapt to shifting market conditions through ongoing learning and model updates, guaranteeing that impairment analysis is reliable and applicable in an unpredictable economic climate.

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1. Introduction

Financial leasing is a vital financing mechanism that enables businesses to access capital-intensive assets without incurring the full cost of ownership upfront. By distributing payments over time, leasing agreements facilitate expansion, enhance liquidity, and offer access to contemporary machinery and technology. But these advantages come with a lot of hazards, especially the potential for asset impairment [8], [9]. When leased assets' estimated recoverable value drops as a result of things like asset deterioration, client defaults, or economic downturns, impairment takes place. A crucial area of attention in financial leasing is impairment analysis since such occurrences not only lower profitability but also raise credit risk for both lessors and lessees [4], [7].

The complex and dynamic character of financial markets cannot be adequately captured by traditional approaches to impairment assessment, which frequently rely on rule-based techniques, static models, or backward-looking financial statistics [1], [2]. A more flexible, data-driven option is provided by machine learning, which can process vast amounts of behavioral and financial data to find hidden patterns and more accurately forecast impairment risks [3], [5], [10], [11]. Leasing firms may transition from reactive to proactive risk management by using machine learning into impairment analysis. This will help them anticipate possible losses, adhere to regulations like IFRS 9 [6], and maintain greater financial stability in an unpredictable economic climate [12].

2. Background & Literature Survey

Comprehensive guidance on financial instrument accounting, including rules for impairment, is provided by the International Financial Reporting Standard (IFRS 9). It places a strong emphasis on the anticipated credit loss (ECL) method, which calls for assessments that look ahead rather than only using past loss data [6], [7]. This standard emphasizes how crucial it is to precisely estimate probable impairments in leased assets in order to maintain financial stability and guarantee appropriate provisioning for financial leasing. Leasing firms must adhere to IFRS 9 in order to reduce credit risk and satisfy regulatory requirements, especially during periods of economic volatility.

Fixed provisioning criteria and statistical models have been major components of traditional impairment detection techniques [1], [2]. To calculate possible losses, these techniques frequently make use of asset depreciation rates, credit scores, and previous performance. These methods are useful in stable settings, but their predictive effectiveness is limited since they are unable to capture non-linear correlations and abrupt changes in the market. Conventional approaches may not be able to promptly identify rising impairment risks in financial leasing portfolios, which comprise a variety of asset types and client profiles.

The use of machine learning and artificial intelligence methods in credit risk and impairment assessments has been investigated recently [3, 5, 8, 10]. By examining intricate financial and behavioral data, methods including supervised learning, neural networks, and ensemble models have demonstrated potential in identifying high-risk clients and forecasting defaults. According to these research, machine learning can offer more adaptive and accurate impairment assessments than conventional techniques, which makes it especially pertinent for portfolios that need ongoing monitoring and risk assessment with an eye toward the future.

Machine learning models such as Random Forest, Gradient Boosting, and deep neural networks have been applied to financial risk assessment with encouraging results [13], [14], [11], [15], [9]. These algorithms can manage high-dimensional features, automatically identify patterns in big datasets, and give decision-makers comprehensible risk scores. Nevertheless, there is a clear lack of application of these sophisticated models to financial leasing portfolios in particular [12]. Leasing assets, in contrast to bank loans, have special features that call for specialized modeling techniques, such as residual value risk and different depreciation trends. By creating machine learning-based impairment analysis techniques tailored for leasing, this study seeks to close that gap by utilizing both historical and behavioral data to produce forecasts that are more accurate.

3. Research Gap

- Static nature of existing models – Current impairment models are largely rule-based or statistical and cannot dynamically adjust to sudden economic or market fluctuations [1], [2].
- Limited focus on leasing-specific contexts – Most research in impairment prediction focuses on banking loans and credit scoring, while financial leasing has unique risk factors that remain underexplored [8], [15].
- Inadequate use of real-time data – Existing models seldom integrate real-time financial indicators, macroeconomic shifts, or customer behavior patterns into impairment prediction [4], [7].
- Overreliance on historical data – Many impairment studies rely solely on past financial records, ignoring forward-looking risk signals that could provide early warnings [9].
- Lack of explainability in AI models – Some machine learning models proposed in finance lack transparency, making them unsuitable for regulatory requirements and stakeholder trust [11], [12].
- Limited integration with IFRS 9 requirements – Few studies address how machine learning-based impairment models can align with International Financial reporting standards (IFRS 9) on expected credit loss [6].
- Insufficient comparative studies – There is limited research comparing the performance of machine learning models against traditional impairment analysis methods in leasing [3], [5].
- Data quality and preprocessing challenges – Impairment datasets in leasing are often incomplete, imbalanced, or inconsistent, yet few studies propose robust preprocessing and feature engineering strategies [13], [14].
- Neglect of asset-specific characteristics – Existing models often treat leasing assets as homogeneous, ignoring variations in asset type, usage, and depreciation that affect impairment risks [9].
- Scarcity of predictive dashboards/decision tools – Research is mostly theoretical, with limited focus on developing practical, user-friendly systems for leasing companies to monitor and act on impairment risks [10], [12].

4. Research Problem

The volatile nature of markets and the rising complexity of client portfolios provide growing problems for the financial leasing industry. Conventional impairment assessment techniques, which mostly rely on past financial data and set statistical principles, frequently aren't able to predict new hazards or adjust to abrupt changes in the economy. It is crucial to investigate more flexible and precise methods of impairment analysis since this restriction exposes leasing organizations to increased credit risk and possible financial losses.

By using vast amounts of behavioral and historical data to find patterns and connections that traditional approaches can miss, machine learning offers a chance to overcome these drawbacks. Leasing businesses can prioritize high-risk accounts, estimate the expected credit loss (ECL) for each asset more precisely, and take preventative action to lessen possible losses by incorporating predictive analytics into impairment assessment. This strategy is especially pertinent in light of IFRS 9 requirements, which place a strong emphasis on risk assessment in the future and call for reliable methods for estimating impairment.

Machine learning has a lot of potential, but its use in financial leasing is still somewhat limited. Leasing-specific portfolios are understudied because the majority of current research has been on credit risk in banking loans or generic financial instruments. Predicting impairments is made more difficult by the distinctive features of leasing assets, which can include different depreciation trends, varied residual values, and a variety of asset kinds. To guarantee that machine learning models offer trustworthy and useful insights specific to leasing portfolios, it is imperative to address these subtleties.

By examining the systematic application of machine learning approaches to impairment analysis in financial leasing, this study aims to close the gap. In addition to increasing prediction accuracy above conventional models, the goal is also to guarantee regulatory compliance, increase decision-making procedures, and assist risk management plans. This study intends to equip leasing firms with tools that are flexible, interpretable, and able to react to changing market conditions by creating a framework that integrates both financial and behavioral data.

5. Objectives

5.1. Main objective

The creation of a machine learning-based system for financial leasing impairment analysis is the main goal of this study. In comparison to conventional statistical models, this method seeks to offer a more precise, flexible, and forward-looking evaluation of possible asset impairments. The system aims to assist lease organizations in limiting financial losses, adhering to legal standards, and making proactive risk management decisions by utilizing vast amounts of historical and behavioral data.

5.2. Specific objectives

A number of specific goals have been developed in order to accomplish the overall goal. Gathering and preprocessing thorough financial leasing data, such as details on leased assets, customer payment patterns, and macroeconomic factors, is crucial first. The basis for successful machine learning modeling will be high-quality, cleansed, and organized data.

The second goal of the study is to pinpoint important behavioral and financial markers that have a major impact on impairment. This entails examining both qualitative behavioral patterns, such as customer risk behavior and market mood, and quantitative financial data, such as payment history, residual asset prices, and credit scores. Gaining insight into these elements will increase the models' capacity for prediction.

Third, in order to forecast the probability of impairment for any leased asset, the study plans to use a range of machine learning models, including supervised and ensemble approaches. To make sure these models appropriately represent the intricate patterns and non-linear correlations present in lease portfolios, they will be trained and verified using historical data.

Fourth, to show the system's efficacy and room for development, it will be compared to conventional impairment analysis techniques. Prediction accuracy, adaptability to shifting market conditions, and IFRS 9 compliance will be the main topics of comparative examination.

Lastly, the study aims to guarantee that the suggested approach complies completely with the impairment requirements of IFRS 9. This entails incorporating risk assessments that look forward, producing outputs that can be explained for regulatory compliance, and giving decision-makers with useful information to help them reduce the financial risks connected to impaired assets.

6. Methodology

6.1. Overall system description

The goal of the research's suggested methodology is to create a reliable machine learning-based system for financial leasing impairment analysis. The system starts by gathering historical leasing data, which includes details about leased assets, payment patterns of lessees, macroeconomic factors, and portfolio attributes. Because of the vast amount of data—which might surpass 180,000 records and contain over ten pertinent attributes per record—careful preparation is necessary to guarantee data completeness, consistency, and quality. In order to increase the machine learning models' dependability, this preprocessing stage entails handling missing values, normalizing numerical features, encoding categorical variables, and resolving any possible class imbalances.

In order to determine the most important behavioral and financial indications that lead to asset impairment, the system will extract and select features after preprocessing. To increase the models' capacity for prediction, feature engineering methods including creating ratios, lagged variables, and trend indicators will be used. By ensuring that the models are both accurate and interpretable, this stage enables analysts to comprehend the factors that influence impairment risk.

Training different machine learning models to forecast the likelihood of asset impairment is the basis of the system. The predicted performance of supervised learning algorithms, such as Random Forest, Gradient Boosting, XGBoost, and Neural Networks, will be tested and evaluated. To verify robustness and dependability, the model will be evaluated using cross-validation methods and metrics including accuracy, precision, recall, F1-score, and area under the ROC curve (AUC). To confirm robustness in challenging circumstances, the models will also be stress-tested under fictitious economic slump scenarios.

Lastly, a reporting and visualization module that displays impairment risk at the portfolio and individual asset levels will be integrated into the system. Leasing managers will be able to prioritize high-risk assets, take proactive steps to reduce risk, and make decisions that comply with the standards of IFRS 9 Expected Credit Loss (ECL) thanks to this module. The process guarantees that the final system is scalable, complies with legal requirements, and can assist in making strategic financial decisions for the leasing industry.

6.2. System diagram

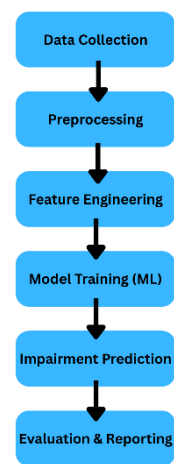


Figure 1.1 System Diagram

6.3. Gantt Chart

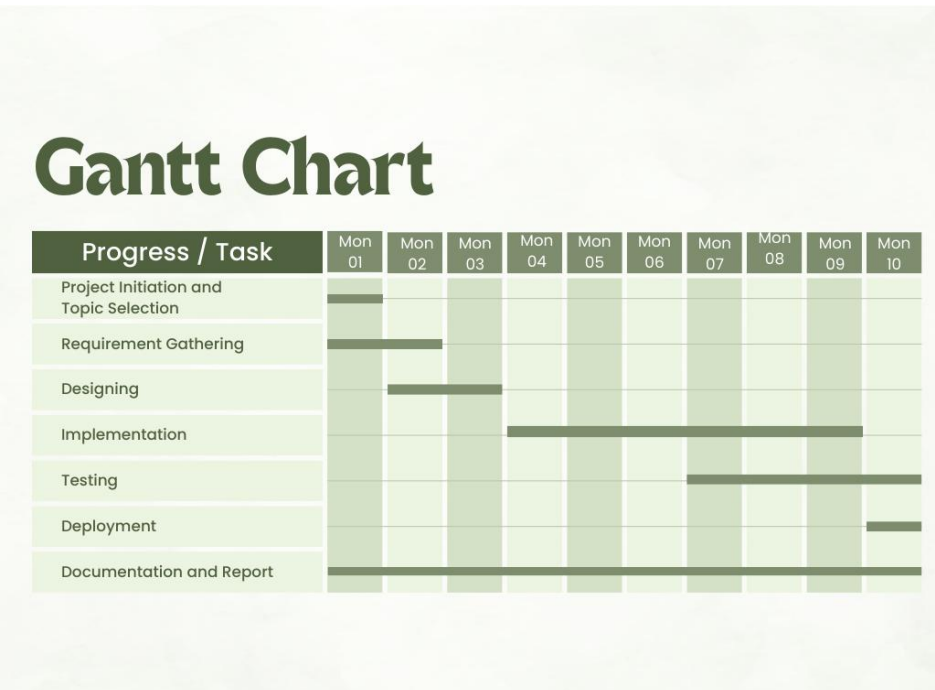


Figure 2.1: Gantt Chart

7. Project Requirements

7.1. Functional requirements

- Import and preprocess leasing data.
- Generate impairment prediction outputs.
- Visualize impairment risk at portfolio and asset levels.
- Provide automated reporting with exportable formats (PDF, Excel, Dashboard).
- Enable real-time model retraining with new data.
- Integrate with existing financial/leasing management systems.
- Provide user authentication and role-based access to system functions.

7.2. Non-functional requirements

- High accuracy and scalability.
- Compliance with financial regulations (IFRS 9).
- Secure data handling and confidentiality.
- System should be highly available (99.9% uptime).
- Provide explainable AI outputs to ensure transparency in predictions.
- Ensure low latency for prediction results (<2 seconds).
- Support multi-language and regional financial standards for broader adoption.

7.3. Expected test cases

- Validate impairment predictions with historical impaired assets.
- Stress test with sudden financial downturn scenarios.
- Cross-validate with multiple ML models for reliability.
- Evaluate system performance under high-volume data input.
- Test data preprocessing accuracy (missing values, outliers, normalization).
- Verify compliance with IFRS 9 reporting standards.
- Test security protocols (encryption, role-based access, data privacy compliance).

8. Tools and Technologies

Table 2.2 Tools and Technologies

Category	Tool / Technology	Purpose / Usage
Frontend Development	React	Building interactive user interfaces and dashboards for leasing data analysis
	HTML5, CSS3, TypeScript	Structuring, styling, and scripting the frontend application
Backend Development	.NET Core	Developing APIs for data access, processing, and integration with ML models
	REST API	Communication between frontend and backend
Machine Learning & Data Science	Python (scikit-learn, TensorFlow, PyTorch)	Data preprocessing, feature extraction, model training and evaluation
	Pandas, NumPy	Data handling, manipulation, and analysis
	Matplotlib, Seaborn	Visualization of data trends and model results
Database Management	SQL Server / MySQL	Storage and querying of leasing datasets
Project Management	Jira	Task tracking, sprint planning, and team collaboration
Version Controlling	Git & GitHub	Version control and collaborative development
Collaboration & Communication	Microsoft Teams	Real-time communication and project discussions
Reporting & Documentation	MS Word	Preparing reports, documentation, and data presentation
Compliance & Standards	IFRS 9 Guidelines	Ensuring compliance with financial reporting standards

9. Budget and Budget Justification

Category	Estimated Cost	Justification
Computing Resources (Local / University Lab)	0	Use existing university lab PCs/servers; no extra cost.
Software & Tools	0	Open-source tools (Python, Scikit-Learn, TensorFlow, Pandas, NumPy, Matplotlib).
Data Preprocessing	0	Research assistant/student effort to clean 180,000+ rows and 10+ columns.
Model Development & Testing	5000	Effort to develop, tune, and validate ML models for impairment prediction.
Statistical & Financial Analysis	0	Exploratory data analysis, IFRS 9 alignment, and feature selection.
Dissemination (Reports / Printing)	1500	Printing, report/thesis preparation, minor publication costs.
Contingency (10%)	500	For any minor unforeseen research-related costs.

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